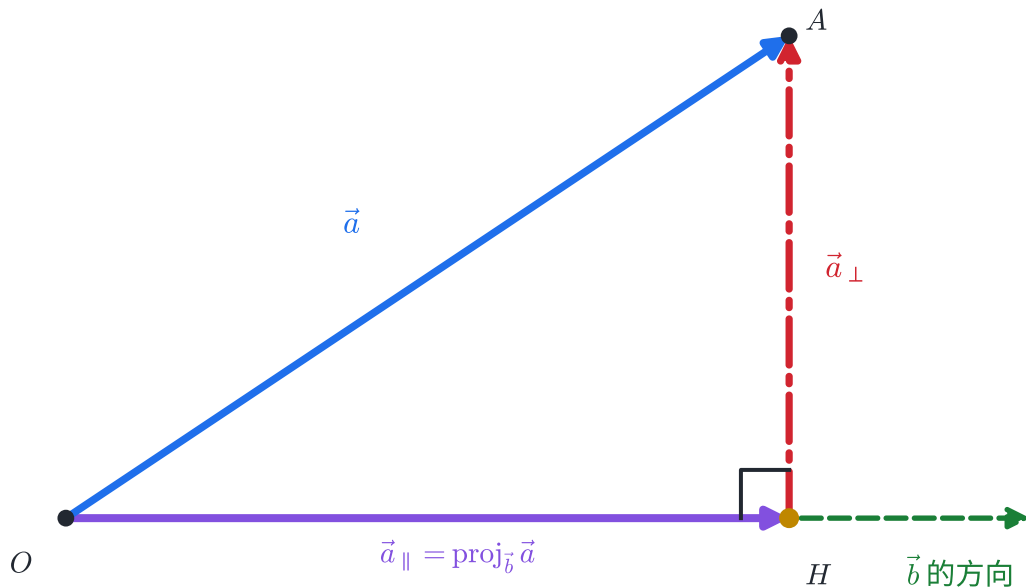


正射影分解：平行分量加上垂直分量

$$\vec{a} = \text{proj}_{\vec{b}} \vec{a} + \vec{a}_{\perp}, \quad \vec{a}_{\perp} \cdot \vec{b} = 0$$



$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}$$